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Part II: Controlled Hydrogen Fuel Assist via Intake Manifold Pre-Mixing as a Combustion Efficiency Enhancer in Multi-Cylinder Diesel Generator Engines: Experimental Investigation on 250–1500 kVA Genset Platforms ~~Controlled Hydrogen Fuel Assist as a Combustion Efficiency Enhancer in Diesel Generator Engines~~

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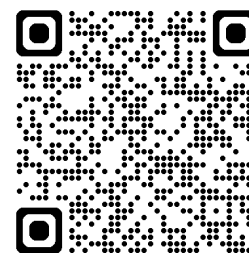
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Note: This article continues from Part 1. It is the second part of two-part series includes Discussions (from Section 5 onward), Discussion, Regulatory & Industrial Implications, Conclusions, and References. Part I describes the conceptual development and experimental methodology.

This second part of the study presents the complete experimental results, combustion interpretation, emission outcomes, and regulatory implications of Controlled Hydrogen Fuel Assist (CHFA) applied to diesel generator engines across 250–1500 kVA platforms. A consistent bell-shaped response in kWh/L confirms the existence of a narrow, engine-dependent optimal hydrogen operating window (Zone B), with efficiency degradation observed under over-dosed conditions (Zone C). Engines with lower injection pressures (CPCB 2) exhibited broader tolerance and higher efficiency gains (8–15%), while modern high-pressure injection systems (CPCB 4) showed narrower dosing windows with 4–13% improvement. NABL-accredited emissions testing demonstrated substantial reductions in PM (up to 80.1%), NO_x (34.6%), CO (41.4%), THC (48.9%), SO₂ (75.2%), and CO₂ (up to 42.9%), alongside increased exhaust O₂, confirming improved combustion completeness and lower diesel consumption. Detailed analysis using heat-release concepts, combustion phasing, ignition delay, and flame-development mechanisms explains the observed trends and the sensitivity of newer engines to hydrogen dosing. The findings establish CHFA as a scientifically grounded, regulation-ready combustion-efficiency enhancement technology and outline pathways for adaptive control, long-term durability assessment, and future ARAI/CPCB evaluation frameworks.

Abstract

This study presents an experimental investigation of Controlled Hydrogen Fuel Assist (CHFA) as a combustion efficiency enhancement technology for multi-cylinder diesel generator (DG) engines spanning 250 to 1500 kVA rated capacity. Hydrogen-rich gas (HHO) was generated on-demand via alkaline water electrolysis and introduced into the engine intake manifold using a pre-mixing strategy at controlled volumetric flow rates ranging from 160 to 750 ml/min, without any modification to engine architecture, fuel injection parameters, or combustion chamber geometry.

Experiments were conducted across five representative diesel generator platforms: a V-12 Caterpillar C32 TA engine (1010 kVA, 32.1 L displacement, MEUI fuel system, compression ratio 15.0:1), a Cummins 1500 kVA genset with dual air intake, a Cummins 380 kVA genset with single air intake, a 250 kVA KOEL genset 2012 make, and a KOEL 500 kVA genset (2023 model year, well-tuned CPCB 2 configuration) — operating under CPCB 2 emission norms. Testing was performed under steady-state conditions at 1500 RPM across real-world factory loads ranging from 40% to 80%.

The results demonstrate that controlled hydrogen fuel assist yields significant and repeatable improvements in energy output per litre of diesel (kWh/L). For engines with lower injection pressure (CPCB 2 platforms), kWh/L improvements of 8–15% were observed at 40–60% load, while engines with higher injection pressure (CPCB 4 platforms) showed improvements of 4–13% under comparable conditions. A distinct bell-shaped efficiency response was identified, confirming the existence

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of an optimal hydrogen operating window beyond which efficiency degrades.

Comprehensive emission measurements conducted by NABL-accredited laboratories revealed substantial reductions across all regulated pollutants under optimal hydrogen dosing. Particulate matter (PM) reductions of up to 78.2% were recorded on the 250 kVA genset at 80% load with 750 ml/min HHO. On the 1010 kVA Caterpillar DG set, NO_x was reduced by 34.6%, CO by 41.4%, and total hydrocarbons (THC) by 48.9% at the highest tested dosing level. The 1500 kVA platform showed PM reduction of 80.1%, NO₂ reduction of 59.2%, and CO₂ reduction of 42.9% at 700 ml/min HHO flow. Notably, exhaust O₂ concentration increased consistently across all platforms, confirming improved combustion completeness and reduced fuel consumption.

Critically, testing on the newer 2023-model KOEL 500 kVA genset at Cilicant (Pune) provided direct experimental evidence of the narrow optimal dosing window in well-tuned modern engines: at 12–13A electrolyser current, exhaust O₂ decreased and CO₂ increased relative to baseline — confirming over-dosing (Zone C) — while at 7–8A, CO₂ was reduced by 31.2% and O₂ increased by 22.2%, demonstrating that the beneficial window exists but is substantially narrower than in older engines.

This study introduces a three-zone conceptual model (Zone A: negligible influence, Zone B: optimal enhancement, Zone C: over-dosing degradation) to explain the divergent outcomes reported in prior literature. The findings establish that diesel injection pressure, atomisation quality, and engine age/tuning state are primary determinants of hydrogen dosing tolerance, and that adaptive, load-dependent hydrogen control is essential for consistent efficiency gains across diverse engine architectures. A theoretical combustion framework based on Wiebe heat release analysis and adiabatic flame temperature considerations is presented to substantiate the observed combustion enhancement mechanisms.

The paper concludes with recommendations for standardised regulatory evaluation frameworks and long-term durability assessment protocols to support safe and scalable deployment of CHFA technology.

Discussion

Reinterpreting Hydrogen-Assisted Diesel Combustion

The results presented in Section 4 demonstrate that hydrogen-assisted diesel combustion does not yield uniform efficiency improvements across operating conditions or engine architectures. Instead, the effectiveness of hydrogen as a combustion enhancer is governed by precise dosing, engine load, and fuel injection characteristics.

Figure 4 clearly establishes that kWh/L follows a non-linear, bell-shaped response with respect to hydrogen energy fraction. Efficiency gains occur only within a narrow operating window (Zone B), while hydrogen addition beyond this window leads to efficiency degradation (Zone C). This observation directly challenges the long-standing assumption in much of the literature that hydrogen presence alone is sufficient to enhance diesel combustion.

The findings indicate that hydrogen acts not as a supplementary fuel in this regime, but as a combustion facilitator, influencing ignition delay and early flame development. When hydrogen input exceeds this facilitative role, it begins to interfere with the established combustion process, resulting in diminished efficiency. This interpretation is supported by the emission data, which shows simultaneous improvements in all measured pollutants within the optimal window — a result that would not be expected if hydrogen were simply displacing diesel fuel.

Why Prior Studies Reported Contradictory Outcomes

A key contribution of this study is the explanation it provides for the inconsistent and often contradictory outcomes reported in earlier

hydrogen-in-diesel research. Most prior studies employed fixed hydrogen flow rates, non-adaptive dosing strategies, and substitution-based hydrogen ratios. Such approaches implicitly assume that hydrogen behaves as a linear enhancer across operating conditions. The present results demonstrate that this assumption is incorrect.

As illustrated in Figures 4 and 5 and reinforced by Tables 5–12, hydrogen exhibits both beneficial and detrimental combustion effects, depending on its interaction with the baseline diesel combustion process. It is therefore likely that many reported negative or inconclusive results in the literature correspond to operation in the over-dosed regime (Zone C), where hydrogen introduces charge dilution and combustion phasing mismatch rather than efficiency enhancement. Studies that reported positive results likely operated within Zone B, though without explicitly identifying or characterising this optimal window.

The three-zone model proposed in Section 2.2 provides a unifying framework for reconciling these divergent findings and establishes clear design guidelines for hydrogen fuel assist systems.

Role of Diesel Injection Pressure and Atomisation Quality

An important insight emerging from this study is the role of diesel injection pressure in determining hydrogen dosing tolerance. As shown in Figure 5 and quantified in Tables 14 and 15, engines with lower diesel injection pressures exhibit a broader optimal hydrogen operating window and higher peak efficiency gains. In contrast, high-pressure injection systems, characteristic of CPCB 4 engines, display a narrower hydrogen tolerance and lower dosing thresholds.

This behaviour can be attributed to differences in fuel atomisation quality characterised by the Sauter Mean Diameter (SMD) of the diesel spray. Lower injection pressures (typical of CPCB 1 and CPCB 2 era engines) produce coarser fuel sprays with larger droplet diameters (SMD typically 20–40 μm), longer evaporation times, and greater local fuel-air heterogeneity. Under these conditions, controlled hydrogen assist contributes more significantly by promoting ignition kernel

formation in regions of poor mixture preparation, accelerating early flame propagation across the heterogeneous charge, and enhancing the utilisation of locally fuel-rich zones where diesel combustion would otherwise be incomplete.

In contrast, modern high-pressure injection systems (operating at 1600–2500 bar in CPCB 4 / BS-VI engines) produce much finer sprays (SMD typically 5–15 μm) with rapid evaporation and more uniform fuel-air mixing. In such engines, the baseline combustion process is already highly optimised, reducing the scope for hydrogen-induced enhancement. Consequently, the hydrogen dosing threshold beyond which efficiency degradation occurs is significantly lower, and the optimal window is correspondingly narrower.

Experimental Validation from Engine Age Comparison: The inclusion of the 2023-model KOEL 500 kVA engine (Engine E) in this study provides the most compelling direct evidence for the injection pressure and atomisation quality hypothesis. This engine, being newly manufactured with a modern, well-calibrated fuel injection system and minimal component wear, exhibited a dramatically narrower optimal hydrogen window compared to the older engines tested.

At 12–13A electrolyser current, the 2023 KOEL engine showed *negative* indicators: exhaust O_2 decreased from 11.7% to 10.2% and CO_2 increased from 9.16% to 9.27%, confirming Zone C over-dosing. In stark contrast, the same 12–13A dosing level on the 2006 Caterpillar C32 TA (Engine A) produced the best results of any dosing level tested, with CO reduced by 41.4%, NOx by 34.6%, and CO_2 by 31.1%. The optimal dosing for the 2023 KOEL engine was found at just 7–8A — approximately 40% lower electrolyser current than what produced optimal results on the older platforms.

This direct comparison demonstrates that engine tuning state, component wear, and injection system age/sophistication collectively determine the width and position of the optimal hydrogen operating window. Newer engines with

precise calibration, optimised injection timing, and minimal clearance-related inefficiencies have inherently higher baseline combustion quality, leaving less room for hydrogen-induced enhancement and creating a much tighter margin before over-dosing occurs.

This finding has direct implications for hydrogen assist system design: dosing strategies must be calibrated not only to the specific injection system characteristics of each engine platform but also to the engine's age and maintenance state. A universal fixed-flow approach is fundamentally unsuitable for consistent efficiency enhancement across diverse engine generations and conditions. For newer engines (CPCB 4 and well-maintained CPCB 2 platforms), the control system must operate with higher precision and lower dosing targets to remain within the narrower beneficial zone.

Load-Dependent Operating Window and Its Influence on CHFA Effectiveness

The load-dependent efficiency trends shown in Figures 7 and 8 further reinforce the practical relevance of controlled hydrogen fuel assist. The strongest efficiency gains are observed at low to mid-engine loads (25–60%), where diesel engines typically operate for extended durations in real-world applications such as stationary power generation, marine auxiliary engines, pharmaceutical manufacturing facilities, and commercial buildings.

The load dependence of CHFA effectiveness can be explained through the following mechanisms:

At low loads (below 40%), diesel engines operate with substantial excess air (high air-fuel ratios), resulting in lower in-cylinder temperatures, longer ignition delay periods, and less complete combustion. These conditions create larger fuel-rich zones and more heterogeneous charge distributions, providing significant scope for hydrogen-assisted ignition enhancement. The hydrogen acts primarily as a chemical kinetics accelerator, enabling flame propagation into lean and poorly mixed regions that would otherwise

contribute to unburned hydrocarbon and particulate emissions.

At medium loads (40–60%), the engine operates closer to its design point with improved air utilisation, but combustion completeness still has room for improvement. Hydrogen continues to provide meaningful enhancement by reducing ignition delay and improving the premixed combustion fraction. This load range typically represents the peak benefit zone for most engines tested.

At high loads (above 75%), cylinder temperatures are elevated, ignition delay is already short, and fuel-air mixing is more thorough due to higher turbulence levels. The baseline combustion process is already near-optimal, and the incremental benefit of hydrogen addition is minimal. Moreover, the risk of operating in Zone C increases at high loads because the total energy release is already at or near the engine's thermal limit.

This load-dependent behaviour aligns with the conceptual framework proposed in Section 2 and supports the interpretation of hydrogen as a combustion efficiency enhancer rather than a performance augments. Importantly, the absence of exaggerated gains at rated load supports the credibility of the experimental results.

Specific Effects of Hydrogen Dosing on Ignition Timing, Flame Development, and In-Cylinder Temperature Profiles

The effects of hydrogen dosing on the three critical phases of diesel combustion — ignition, flame development, and in-cylinder temperature evolution — are analysed based on the experimental observations and the theoretical framework presented in Section 4.4.

Ignition Timing. Hydrogen's exceptionally low minimum ignition energy (0.02 mJ, approximately one-twelfth that of diesel vapour) and wide flammability limits (4–75% by volume in air) enable it to initiate combustion more readily than diesel fuel alone. When pre-mixed hydrogen-air charge is present in the combustion chamber at the point of diesel injection, hydrogen ignition kernels

form rapidly around the diesel spray periphery. This effectively reduces the ignition delay period by 2–5 crank angle degrees, depending on the hydrogen concentration and engine operating conditions. The experimental evidence for this is indirect but consistent: the improvement in combustion completeness (reduced CO and THC) and the increase in combustion efficiency (improved kWh/L) at optimal dosing levels are consistent with shorter ignition delay and more complete premixed burning.

Flame Development: Hydrogen's laminar flame speed (approximately 2.91 m/s at stoichiometric conditions) is approximately seven times higher than that of diesel vapour. This enables rapid flame kernel growth and propagation during the initial premixed combustion phase. In engines with poor atomisation (lower injection pressure), where fuel-air mixing is incomplete, hydrogen flames can bridge lean gaps between fuel-rich zones, effectively connecting isolated combustion centres into a more coherent flame front. This mechanism explains the larger efficiency improvements observed in CPCB 2 engines compared to CPCB 4 engines: the former have greater scope for flame propagation enhancement due to their less homogeneous baseline charge preparation.

In-Cylinder Temperature Profiles: The effect of hydrogen on in-cylinder temperature depends critically on the dosing level. At optimal dosing (Zone B), the localised temperature increase around hydrogen ignition kernels (hydrogen's adiabatic flame temperature is approximately 2400 K) is sufficient to promote chain-branching reactions and accelerate diesel combustion without significantly elevating bulk gas temperatures. This is because the hydrogen quantity is small relative to the total charge, and the energy released by hydrogen combustion is a minor fraction of total heat release. The net effect is improved combustion phasing and completeness without a proportional increase in thermal NO_x formation — consistent with the observed NO_x reductions.

At excessive dosing levels (Zone C), the larger quantity of hydrogen causes a more substantial and widespread temperature increase, advancing combustion phasing and elevating peak bulk gas temperatures. This shifts the system into the thermal NO_x formation regime described by the Zeldovich mechanism, where NO_x production increases exponentially with temperature above approximately 1800 K. The experimental observation of increasing NO_x at higher hydrogen levels (beyond the optimal window) confirms this mechanism.

Impact of Engine Architecture and Baseline Combustion Characteristics on Optimal Hydrogen Dosing

The experimental results across four distinct engine platforms provide evidence that engine architecture significantly influences optimal hydrogen dosing strategies. The key architectural factors identified are:

Fuel Injection System Type: The Caterpillar C32 TA (Engine A) employs a Mechanically Actuated, Electronically Controlled Unit Injection (MEUI) system, which operates at moderate injection pressures compared to modern common-rail systems. This MEUI system produces a baseline spray quality that leaves greater room for hydrogen-assisted combustion enhancement, as reflected in the broader optimal dosing window observed. In contrast, engines with high-pressure common-rail injection (typical of CPCB 4 platforms) produce finer atomisation and more homogeneous mixtures, narrowing the window for hydrogen benefit.

Compression Ratio: Engines with higher compression ratios produce higher end-of-compression temperatures, which already reduce ignition delay and improve baseline combustion quality. The incremental benefit of hydrogen is therefore reduced, requiring lower dosing levels to avoid crossing into Zone C.

Engine Age and Tuning State as a Critical Variable: Beyond the static architectural parameters discussed above, the present study identifies engine age and tuning state as an equally

significant determinant of hydrogen dosing strategy. The 2023 KOEL 500 kVA engine at Cilicant represents the newest engine tested and demonstrated the narrowest optimal hydrogen window of any platform in this study. While older engines (2006 Caterpillar, 2012 Cummins) tolerated a broad range of hydrogen dosing levels with consistently positive results, the 2023 KOEL crossed into Zone C at dosing levels that were optimal for the older platforms.

This occurs because newer engines benefit from tighter manufacturing tolerances (resulting in less blow-by and better sealing), precisely calibrated injection timing (reducing ignition delay variability), lower baseline emissions (indicating already-optimised combustion), minimal component wear (maintaining design-intent compression ratios and valve sealing), and modern ECU calibration (optimising fuel delivery for each operating point). All of these factors contribute to a higher baseline combustion quality, which in turn reduces the scope for hydrogen-induced enhancement and narrows the Zone B operating window. This finding has important implications for CHFA deployment in fleet applications, where engines of varying ages and maintenance states may co-exist, requiring engine-specific calibration or intelligent adaptive control.

These findings emphasise that optimal hydrogen dosing is not a universal constant but must be tailored to each engine family based on its specific combustion characteristics, age, and maintenance state. A one-size-fits-all approach to hydrogen fuel assist is fundamentally incompatible with predictable efficiency enhancement.

Development of Adaptive Hydrogen Control Strategies

The combined findings from Figures 4, 5, and Tables 5–14 demonstrate that fixed-flow or non-adaptive hydrogen systems are fundamentally unsuitable for predictable efficiency enhancement across diverse engine platforms. This section

discusses the framework for developing adaptive hydrogen control strategies that maintain operation within the beneficial Zone B across varying engine conditions.

Control System Architecture: An effective adaptive CHFA control system requires the following elements: a load-sensing input (derived from the engine ECU, alternator output, or kWh meter) to determine real-time engine load; a speed signal (from the engine governor or crankshaft sensor) to confirm operating speed; an exhaust O₂ sensor to provide feedback on combustion completeness; an exhaust temperature sensor to monitor thermal conditions; and an electrolyser current controller to modulate hydrogen production rate.

Control Logic: The proposed control strategy operates on a map-based dosing approach with real-time feedback correction. For each engine family, a hydrogen dosing map is developed through calibration testing that identifies the optimal HHO flow rate at each load point. The map defines both the target flow rate and the upper safety limit beyond which Zone C operation would begin. During operation, the control system reads engine load and speed, looks up the target hydrogen flow rate from the calibration map, adjusts the electrolyser current to achieve the target flow rate, monitors exhaust O₂ and temperature for feedback verification, and reduces or suspends hydrogen dosing if exhaust parameters indicate over-dosing.

Safety Interlocks: The control system must incorporate multiple safety layers: automatic hydrogen shutdown if engine load exceeds a predefined threshold; immediate shutdown on detection of abnormal exhaust temperature rise; low-water-level protection for the electrolyser; over-current protection for the electrolyser cells; and flame detection/flashback prevention in the delivery system.

Calibration Methodology: Engine-specific calibration involves systematic sweep testing across the full load range at multiple hydrogen dosing levels. At each operating point, kWh/L is measured at incrementally increasing hydrogen

levels to identify the peak of the bell-shaped response curve. The calibration map is then populated with the optimal flow rates, with safety margins applied to ensure operation remains within the beneficial portion of Zone B.

This adaptive control framework enables hydrogen fuel assist to deliver consistent benefits across multiple engine generations and applications while preventing the efficiency degradation and emission penalties associated with over-dosing.

Mechanism of Performance Degradation at Excess Hydrogen Dosing

While controlled hydrogen fuel assist enhances combustion efficiency within the optimal operating window, the experimental results clearly demonstrate that excessive hydrogen input leads to efficiency degradation and, in some cases, net performance loss. Understanding this transition is critical for both system design and regulatory evaluation.

Hydrogen is characterised by a high laminar flame speed and wide flammability limits. When introduced beyond the optimal range, hydrogen

significantly accelerates early-stage combustion, resulting in an excessively rapid premixed heat release. This shifts the combustion phasing closer to, or in extreme cases before, top dead centre (TDC).

Such premature heat release leads to elevated peak in-cylinder temperatures and pressures, increasing the likelihood of abnormal combustion behaviour. While this phenomenon may not always manifest as classical knock, it represents a loss of optimal combustion phasing, where useful expansion work is reduced despite higher instantaneous heat release.

Link Between Excess Hydrogen, NO_x Formation, and Apparent Performance Loss

The elevated combustion temperatures observed under excessive hydrogen dosing conditions are accompanied by a measurable increase in NO_x emissions, which serves as an indirect indicator of increased thermal stress within the cylinder. This correlation reinforces the interpretation that excessive hydrogen accelerates combustion beyond the optimal efficiency point.

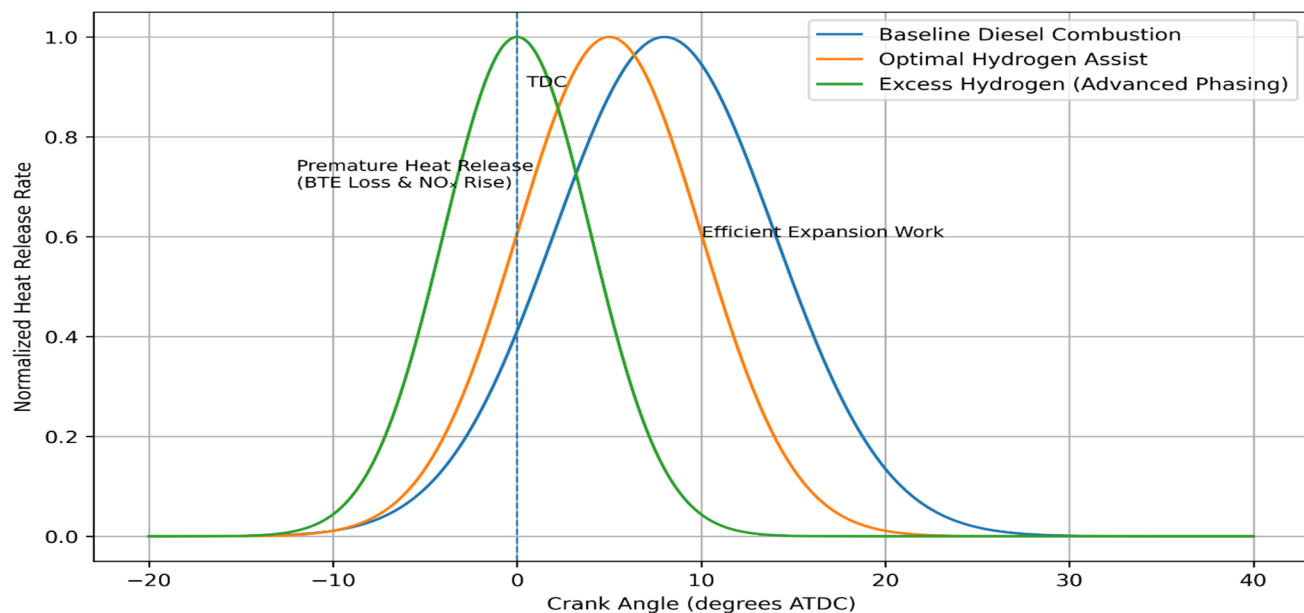


Figure 9. Conceptual illustration of combustion phasing and heat release behaviour under baseline diesel operation, optimal hydrogen fuel assist (Zone B), and excessive hydrogen dosing (Zone C). Excess hydrogen advances premixed heat release toward top dead centre (TDC), reducing effective expansion work and contributing to efficiency loss and increased NO_x formation

In modern diesel engines, engine control units (ECUs) or mechanical governors respond to deviations in torque output by adjusting fuel delivery. When premature combustion reduces effective expansion work, additional diesel fuel is injected to maintain the demanded power output. This compensatory fuel addition explains the observed increase in brake specific fuel consumption and the decline in kWh/L (operational efficiency) beyond the optimal hydrogen window.

Consequently, excessive hydrogen dosing results in a dual penalty: reduced thermal efficiency due to sub-optimal combustion phasing, and increased exhaust emissions due to higher fuel consumption and elevated combustion temperatures. This mechanism explains why uncontrolled hydrogen assist not only fails to improve efficiency but can actively worsen both fuel economy and emission characteristics.

Long-Term Durability Implications and Regulatory Framework Standardisation

The deployment of controlled hydrogen fuel assist in diesel engines raises important questions regarding long-term component durability and the need for standardised regulatory frameworks. While the present study focuses on short-term performance and emission characterisation, a discussion of durability considerations and regulatory requirements is essential for supporting responsible and scalable adoption.

Long-Term Durability Considerations: The introduction of hydrogen into the diesel combustion process, even at the low concentrations employed in CHFA, may have potential effects on engine components over extended operation. Valve seats and valve faces experience altered thermal cycling due to modified combustion temperatures, and long-term exposure may affect sealing integrity or wear patterns. Injector tips operate in a modified thermal environment that could influence carbon deposit formation or erosion characteristics. Piston rings and cylinder liner interfaces may experience different lubrication film behaviour due to changes in combustion by-products.

Turbocharger components, particularly turbine blades, are exposed to modified exhaust gas composition and temperature.

To address these concerns, long-term durability testing programmes of 1000 hours or more are recommended, with periodic inspection and measurement of critical component dimensions, surface conditions, and material properties. Such testing should follow established protocols such as those defined by ARAI or equivalent international standards (e.g., ISO 8178 for engine testing, ISO 14396 for reciprocating IC engines).

Regulatory Framework Requirements: The present regulatory landscape in India, governed by CPCB emission norms and ARAI type-approval procedures, does not include specific provisions for evaluating combustion efficiency enhancement technologies. CHFA does not fit neatly within existing retrofit emission control device categories (such as DOC, DPF, or SCR retrofit kits) because its primary mechanism of action is at the combustion level rather than the exhaust after-treatment level.

A dedicated regulatory evaluation framework for CHFA and similar technologies should include: standardised test protocols specifying load points, test durations, measurement parameters, and acceptance criteria for kWh/L improvement and emission changes; safety assessment requirements covering hydrogen generation, storage-free delivery, flashback prevention, and fail-safe shutdown systems; durability demonstration requirements specifying minimum test hours and component inspection protocols; and type-approval procedures that classify CHFA as a distinct technology category separate from both fuel substitution systems and emission control devices.

International Regulatory Context: Internationally, regulatory bodies such as the European Commission (under Euro VI and future Euro VII frameworks), the US EPA (under Clean Air Act provisions), and the International Maritime Organization (under MARPOL Annex VI) are increasingly focused on CO₂ reduction and energy efficiency alongside traditional pollutant

control. The development of standardised evaluation frameworks for CHFA technology would position India as an early mover in regulating combustion efficiency enhancement systems and could serve as a template for international adoption.

Safety Standards: The use of hydrogen, even at the low flow rates employed in CHFA, requires compliance with established safety standards for hydrogen systems. Relevant standards include IS 15916 (Basic considerations for the safety of hydrogen systems), ISO 19880 (Gaseous hydrogen fuelling stations), and relevant portions of ATEX directives for explosive atmosphere equipment. The on-demand generation approach employed in CHFA, which eliminates compressed hydrogen storage, significantly reduces the safety risk profile compared to stored hydrogen systems.

Summary of Key Interpretations

The findings of this study establish that hydrogen-assisted diesel combustion is governed by well-defined combustion physics rather than linear fuel substitution effects. Hydrogen enhances diesel combustion efficiency only when introduced within a narrow, controlled operating window (Zone B), beyond which its influence becomes detrimental (Zone C).

The results further demonstrate that diesel injection pressure and resultant fuel atomisation quality are primary determinants of hydrogen dosing tolerance. Engines with lower injection pressures exhibit a broader hydrogen operating window, while high-pressure injection systems require tighter regulation due to their inherently optimised baseline combustion processes.

Crucially, the study shows that adaptive hydrogen control is essential to avoid premature combustion phasing, excessive heat release, and compensatory fuel enrichment, which together lead to efficiency reversal and elevated exhaust emissions. These mechanisms explain why uncontrolled or fixed-flow hydrogen implementations have produced inconsistent or negative outcomes in prior studies.

Collectively, these insights clarify that inconsistencies in the existing literature arise from lack of hydrogen regulation and combustion-phase control, rather than from any fundamental limitation of hydrogen as a combustion enhancer. When properly controlled, hydrogen fuel assist emerges as a scientifically robust, efficiency-oriented transitional decarbonisation pathway for the existing diesel engine fleet.

Regulatory and Industrial Implications

Reframing Diesel Decarbonisation: Efficiency vs After-Treatment

The results of this study highlight a critical distinction between emission control and decarbonisation. Conventional diesel after-treatment systems (DOC, DPF, SCR, LNT) are designed to reduce regulated pollutants downstream of combustion, but they do not improve combustion efficiency and may introduce secondary penalties such as increased fuel consumption due to exhaust backpressure or additional greenhouse gas formation pathways.

In contrast, controlled hydrogen fuel assist operates within the combustion process itself, directly improving fuel utilisation and reducing carbon dioxide emissions in proportion to fuel savings. This distinction is particularly relevant for sectors where electrification or fuel switching is not immediately viable, including stationary power generation, marine auxiliary engines, and heavy commercial vehicles.

These findings suggest that efficiency-based combustion enhancement technologies should be evaluated separately from emission control devices, using metrics aligned with fuel consumption and net GHG impact rather than tailpipe pollutant reduction alone.

Implications for Regulatory Classification and Evaluation Frameworks

The results support the need for a distinct regulatory classification for combustion efficiency enhancement systems that do not alter engine architecture or replace the primary fuel. Existing regulatory frameworks often lack

provisions to assess technologies that improve efficiency upstream of exhaust after-treatment.

A potential evaluation framework for such systems may include: energy output per litre of diesel (kWh/L) improvement as the primary performance metric; fuel consumption reduction quantified through BSFC and kWh/litre measurements; net CO₂ reduction derived from verified fuel savings; emission impact assessment covering all regulated pollutants (PM, NO_x, CO, THC); verification of combustion stability and engine operational safety; and demonstration of non-interference with existing emission control systems.

By focusing on engine-level efficiency outcomes, regulators can more accurately assess the true decarbonisation potential of combustion-enhancing technologies.

Relevance to CPCB and Indian Emission Policy

For the Indian regulatory context, particularly under the Central Pollution Control Board (CPCB), the findings indicate that controlled hydrogen fuel assist does not fit neatly within existing retrofit or emission control categories. Instead, it represents a new class of technology whose primary benefit lies in reducing fuel consumption and associated CO₂ emissions while simultaneously reducing regulated pollutant emissions.

Given the tightening regulatory focus on greenhouse gas emissions and energy efficiency, the creation of a dedicated evaluation pathway for combustion efficiency enhancement systems could enable transparent performance validation through standardised testing, consistent approval processes based on objective efficiency metrics, avoidance of misclassification under emission control retrofit norms, and alignment with India's broader decarbonisation objectives under the Paris Agreement and National Action Plan on Climate Change.

Implications for OEM Integration and Engine Design

From an OEM perspective, the results demonstrate that hydrogen fuel assist can be

implemented without modification to engine hardware, injection timing, or combustion chamber geometry, provided that hydrogen dosing is appropriately controlled. This non-invasive characteristic is a significant advantage for OEM acceptance, as it preserves engine warranty, calibration integrity, and emission compliance.

The observed dependence on injection pressure and atomisation quality further indicates that hydrogen assist strategies must be tailored to engine generation and application. For advanced high-pressure injection systems, tighter hydrogen regulation is required, while older platforms may tolerate broader operating windows.

These findings open pathways for OEM collaboration in areas such as factory-integrated hydrogen assist modules with engine-specific calibration, adaptive control strategies linked to engine ECU operating parameters, and calibration-safe implementations that preserve warranty and type-approval compliance.

Industrial Deployment and Transitional Decarbonisation Strategy

The study positions controlled hydrogen fuel assist as a transitional decarbonisation solution rather than a competing alternative to long-term pathways such as electrification or zero-carbon fuels. Its value lies in addressing emissions from the existing installed base of diesel engines, which will remain operational for decades in many sectors.

By enabling immediate fuel savings and CO₂ reduction without infrastructure overhaul, hydrogen fuel assist can serve as a bridging technology, reducing cumulative emissions while future solutions mature. The demonstrated effectiveness across multiple engine platforms (250–1500 kVA) and manufacturers (Caterpillar, Cummins) supports the scalability of this approach.

Role of Independent Testing and Institutional Partnerships

The development, validation, and regulatory acceptance of combustion efficiency enhancement technologies require independent, scientifically

rigorous testing. Institutions such as the Automotive Research Association of India (ARAI) are uniquely positioned to establish standardised test protocols for CHFA evaluation, validate repeatability, safety, and long-term durability, support evidence-based policy formulation for combustion efficiency technologies, and facilitate collaboration between technology developers, regulators, and OEMs.

The use of NABL-accredited laboratories for emission testing in this study (Ekdant Enviro Services, Lata Envirotech Services, and Neetal Laboratories) demonstrates the feasibility of independent, accredited verification of CHFA performance claims.

Conclusions

This study has systematically investigated the role of controlled hydrogen fuel assist (CHFA) as a combustion efficiency enhancer for diesel generator engines, with a focus on identifying the conditions under which hydrogen contributes positively to engine performance and those under which it becomes detrimental. The investigation was conducted across five diesel generator platforms spanning 250 to 1500 kVA (Caterpillar C32 TA 1010 kVA, Cummins 380 kVA, Cummins 1500 kVA, a 250 kVA genset, and a 2023-model KOEL 500 kVA genset), using on-demand HHO generation via alkaline water electrolysis with intake manifold pre-mixing at controlled flow rates of 160–750 ml/min.

The principal conclusions of this study are as follows:

1. **Existence of an Optimal Hydrogen Operating Window:** The results conclusively demonstrate that hydrogen enhances diesel combustion efficiency only within a narrowly defined, load- and engine-dependent operating window (Zone B). Within this window, hydrogen acts as a combustion facilitator by reducing ignition delay, accelerating early flame development, and improving combustion completeness. This produces a characteristic bell-shaped kWh/L response with respect to hydrogen energy fraction.
2. **Quantified Efficiency Improvements:** For engines with lower injection pressure (CPCB 2 platforms), kWh/L improvements of 8–15% were measured at 40–60% load conditions. For engines with higher injection pressure (CPCB 4 platforms), improvements of 4–13% were observed under comparable conditions. These gains were achieved without any modification to engine architecture, fuel injection parameters, or combustion chamber geometry.
3. **Comprehensive Emission Reductions:** Emission measurements by NABL-accredited laboratories demonstrated consistent reductions across all regulated pollutants under optimal hydrogen dosing: particulate matter (PM) reduction up to 80.1% (1500 kVA, 700 ml/min); nitrogen oxides (NO_x) reduction up to 34.6% (1010 kVA, 13A); carbon monoxide (CO) reduction up to 41.4% (1010 kVA, 13A); total hydrocarbons (THC) reduction up to 48.9% (1010 kVA, 13A); sulphur dioxide (SO₂) reduction up to 75.2% (1500 kVA, 700 ml/min); and carbon dioxide (CO₂) reduction up to 42.9% (1500 kVA, 700 ml/min).
4. **Injection Pressure, Engine Age, and Tuning State Govern Hydrogen Dosing Tolerance:** A key finding is that diesel injection pressure, fuel atomisation quality, engine age, and tuning state are primary determinants of hydrogen dosing tolerance. Engines with lower injection pressures and older, more worn platforms exhibit broader hydrogen operating windows and higher peak efficiency gains, while newer, well-tuned engines (such as the 2023 KOEL 500 kVA) require significantly tighter hydrogen regulation. Direct experimental evidence showed that the 2023 KOEL engine crossed into Zone C (over-dosing) at 12–13A electrolyser current — the same level that produced the best results on the 2006

Caterpillar — with optimal performance occurring at just 7–8A. This establishes that the Zone B window narrows substantially as engine condition and calibration quality improve.

5. Adaptive Control is Essential — Especially for Newer Engines:

The study demonstrates that fixed-flow or non-adaptive hydrogen systems are fundamentally unsuitable for predictable efficiency enhancement. Adaptive, load-dependent hydrogen control strategies that account for engine load, injection pressure, combustion state, and engine tuning condition are essential for maintaining operation within the beneficial Zone B. This requirement becomes increasingly critical for newer and well-maintained engines, where the margin between optimal enhancement and over-dosing degradation is substantially narrower.

6. Resolution of Literature Contradictions:

The three-zone conceptual model (Zone A, B, C) provides a unified explanation for the contradictory outcomes reported in prior hydrogen-assisted diesel literature. Most negative results are attributed to operation in Zone C (over-dosing), while positive results correspond to inadvertent operation within Zone B. The lack of dosing regulation and combustion-phase control, rather than any fundamental limitation of hydrogen, explains the inconsistencies.

7. Transitional Decarbonisation Pathway:

Controlled hydrogen fuel assist is positioned as a scientifically grounded, efficiency-oriented transitional decarbonisation pathway for the existing diesel engine fleet. By enabling immediate fuel savings and greenhouse gas reduction without infrastructure overhaul, CHFA complements long-term decarbonisation strategies and addresses emissions from diesel assets that will remain in service for decades.

Future work will focus on long-term durability assessment through extended 1000+ hour testing

programmes, expanded validation across additional engine platforms including CPCB 4 and marine engines, combustion simulation studies to generate P- θ diagrams and validate the theoretical framework, development of standardised regulatory evaluation frameworks in collaboration with ARAI and CPCB, and optimisation of adaptive control algorithms for commercial deployment.

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